Copy-Paste Augmentation Report (Seed 123)

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1 Objective

This report has 3 main objectives:

- Summarize what copy-paste experiments are currently being executed.
- Quantify how each selected copy-paste configuration affects detection performance.
- Clearly state the current problems observed from the latest seed-123 results.

2 Research Summary

Current focus: We are analyzing **how copy-paste augmentation configuration choices affect AP metrics**.

The two completed experiments used in this report are:

- **Exp 15 (MinneApple):** ratio sweep with score-guided selection and fixed `same_image_only=true`.
- **Exp 17 (Global Wheat Head):** A/B test of `same_image_only` on/off under random selection.

Important scope note: This report intentionally uses **seed 123 only** as requested. Therefore, conclusions should be treated as directional and not final until multi-seed confirmation.

3 What Experiments I Am Doing

3.1 Experiment 15: Dataset Ratio Sweep (MinneApple)

Goal: Check how the number of generated augmented images (`dataset_ratio`) changes performance.

Main controlled setup:

- Dataset: MinneApple.
- Selection mode: score-guided (`use_score_guidance=true`).
- `same_image_only=true`.
- Ratios tested: {0.5, 1.0, 2.0, 3.0}.
- Other copy-paste settings fixed: min/max pasted objects (2, 8), reuse caps (3, 3), overlap avoidance enabled.

3.2 Experiment 17: Same-Image-Only Flag Check (Global Wheat Head)

Goal: Decide whether `same_image_only` should be enabled.

Main controlled setup:

- Dataset: Global Wheat Head.
- Selection mode: random (`use_score_guidance=false`).
- `dataset_ratio=2.0` fixed.
- Compare two conditions only:
  - `same_image_only=true`
  - `same_image_only=false`

4 Config Impact on Performance (Seed 123)

4.1 Exp : Effect of `expansion_ratio` on MinneApple

Baseline on original test set (before copy-paste):

$$\text{COCO\_AP} = 0.3822, \quad \text{COCO\_AP50} = 0.7077, \quad \text{COCO\_AP75} = 0.3741$$

Table 1. Exp 15 (Seed 123): Ratio sweep vs baseline

Setting	COCO_AP	ΔAP	COCO_AP50	COCO_AP75	AP_small	AP_large
Baseline	0.3822	0.0000	0.7077	0.3741	0.2440	0.1386
Ratio = 0.5	0.3541	-0.0280	0.6513	0.3512	0.2140	0.2931
Ratio = 1.0	0.3560	-0.0261	0.6466	0.3581	0.2154	0.3376
Ratio = 2.0	0.3533	-0.0289	0.6432	0.3576	0.2093	0.3366
Ratio = 3.0	0.3470	-0.0352	0.6288	0.3467	0.2063	0.2366

Observation (Exp 15):

- All tested ratios reduce overall COCO_AP relative to baseline.
- The least drop appears at ratio 1.0, but it is still negative.
- AP_large increases strongly while AP and AP_small drop, indicating a trade-off rather than a true global gain.

4.2 Exp 17: Effect of `same_image_only` on Global Wheat Head

Baseline on original test set (before copy-paste):

$$\text{COCO\_AP} = 0.5090, \quad \text{COCO\_AP50} = 0.9259, \quad \text{COCO\_AP75} = 0.4901$$

Table 2. Exp 17 (Seed 123): same_image_only on/off vs baseline

Setting	COCO_AP	ΔAP	COCO_AP50	COCO_AP75	AP_small	AP_large
Baseline	0.5090	0.0000	0.9259	0.4901	0.1346	0.5525
same_image_only = true	0.4989	-0.0101	0.9188	0.4803	0.1055	0.5487
same_image_only = false	0.5014	-0.0075	0.9173	0.4866	0.0932	0.5585

Observation (Exp 17):

- Both settings reduce overall COCO_AP compared with baseline.
- `same_image_only=false` is better on overall AP and AP75.
- `same_image_only=true` is better on AP_small, but worse on overall AP.

5 Multi-Seed Inspection, Configuration, and Results (Exp 12, 13, 19)

5.1 Experiment 12 (MinneApple): with-score vs without-score (3 seeds)

Inspection:

- Purpose: compare score-guided selection against random selection under the same copy-paste setup.
- Seeds: {123, 456, 789}.
- Main finding: both conditions are below baseline on mean ΔCOCO_AP, but score-guided is less negative.

Table 3. Exp 12 configuration summary

Field	Value
Dataset	MinneApple
Seeds	123, 456, 789
Conditions	with_score, without_score

Table 4. Exp 12 COCO results (3-seed aggregate means)

Setting	COCO_AP	ΔAP	COCO_AP50	COCO_AP75	AP_small	AP_large
Baseline	0.3844	0.0000	0.7065	0.3825	0.2497	0.1069
with_score	0.3739	-0.0105	0.6873	0.3722	0.2413	0.7288
without_score	0.3677	-0.0167	0.6677	0.3703	0.2259	0.2950

Exp 12 recommendation: `with_score` (advantage in mean ΔCOCO_AP: +0.0062).

5.2 Experiment 13 (Global Wheat Head): with-score vs without-score (3 seeds)

Inspection:

- Purpose: replicate the same comparison as Exp 12 on Global Wheat Head.
- Seeds: {123, 456, 789}.
- Main finding: both conditions are below baseline, and random selection (without-score) is slightly better than with-score.

Table 5. Exp 13 configuration summary

Field	Value
Dataset	Global Wheat Head
Seeds	123, 456, 789
Conditions	with_score, without_score

Table 6. Exp 13 COCO results (3-seed aggregate means)

Setting	COCO_AP	ΔAP	COCO_AP50	COCO_AP75	AP_small	AP_large
Baseline	0.5070	0.0000	0.9223	0.4907	0.1365	0.5499
with_score	0.4837	-0.0233	0.9045	0.4549	0.0985	0.5308
without_score	0.4857	-0.0213	0.9048	0.4556	0.1182	0.5287

Exp 13 recommendation: `without_score` (mean ΔCOCO_AP advantage: +0.0020 over with-score).

5.3 Experiment 19 (MinneApple): score-alpha sweep (1, 3, 5) with 3 seeds

Inspection:

- Purpose: test sensitivity of score-guided sampling strength via `score_alpha` in {1, 3, 5}.
- Seeds: {123, 456, 789}.
- Main finding: all tested alpha values remain below baseline on mean ΔCOCO_AP; alpha=1 is the least negative.

Table 7. Exp 19 configuration summary

Field	Value
Dataset	MinneApple
Seeds	123, 456, 789
Condition family	score-guided only (`use_score_guidance=true`)
Sweep variable	`score_alpha` in {1.0, 3.0, 5.0}

Table 8. Exp 19 COCO results (3-seed aggregate means)

Setting	COCO_AP	ΔAP	COCO_AP50	COCO_AP75	AP_small	AP_large	Rank
Baseline	0.3844	0.0000	0.7065	0.3825	0.2497	0.1069	–
alpha = 1.0	0.3720	-0.0124	0.6852	0.3711	0.2343	0.6370	1
alpha = 3.0	0.3652	-0.0192	0.6656	0.3654	0.2282	0.4322	2
alpha = 5.0	0.3619	-0.0225	0.6658	0.3602	0.2224	0.3243	3

Exp 19 recommendation: prefer `score_alpha=1.0` among tested values.

6 Current Problems Seen From These Results

6.1 Problem 1: Copy-paste strength (dataset ratio) is not giving net AP gains in Exp 15

Even the best tested ratio still underperforms baseline in overall COCO_AP. This suggests that simply increasing or decreasing generated sample count is not enough to improve generalization on MinneApple under current paste quality and placement settings.

6.2 Problem 2: The `same_image_only` switch changes metric trade-offs, but does not solve the AP drop

In Exp 17, turning `same_image_only` off gives slightly better overall AP, but both on/off remain below baseline. This means the flag matters, but it is not the root cause of the performance gap.

6.3 Problem 3: Metric behavior is inconsistent across object scales

From Exp 15 and Exp 17, some settings help one slice (for example AP_large or AP_small) while hurting overall AP. This indicates augmentation is changing the training distribution in a way that is not uniformly beneficial.

7 Practical Conclusion (Seed 123 Only)

- For Exp 15, no tested `dataset_ratio` improves overall AP over baseline.
- For Exp 17, if forced to pick one option from this seed, `same_image_only=false` is preferable for overall AP.
- Current bottleneck is likely not a single scalar hyperparameter; it is likely related to paste realism, localization quality after paste, or distribution mismatch introduced by synthetic samples.

Appendix: Raw Numbers Used (Seed 123)

Exp 15 sources:

- Baseline: `results/15_find_dataset_ratio_minneapple/seed_123/Step_2_Train_and_Evaluate_BASELI`
- Ratios: `results/15_find_dataset_ratio_minneapple/seed_123/Step_5_Train_and_Evaluate_Model_o`

Exp 17 sources:

- Baseline: `results/17_check_same_image_only_flag/seed_123/Step_2_Train_and_Evaluate_BASELINE_`
- same_image_only on/off: `results/17_check_same_image_only_flag/seed_123/Step_5_Train_and_Evalu`

Exp 12 source:

- Summary: `results/12_multiseed_score_vs_noscore_copy_paste_exp/summary_score_vs_without_sco`

Exp 13 source:

- Summary: `results/13_multiseed_score_vs_noscore_copy_paste_exp_global_wheat_head/summary_sc`

Exp 19 source:

- Summary: `results/19_alpha_sweep_multiseed_cached/summary_alpha_sweep_3seeds.json`